INCH - POUND

MIL-PRF-27/1F
13 July 2012
SUPERSEDING
MIL-PRF-27/1E
23 July 1990

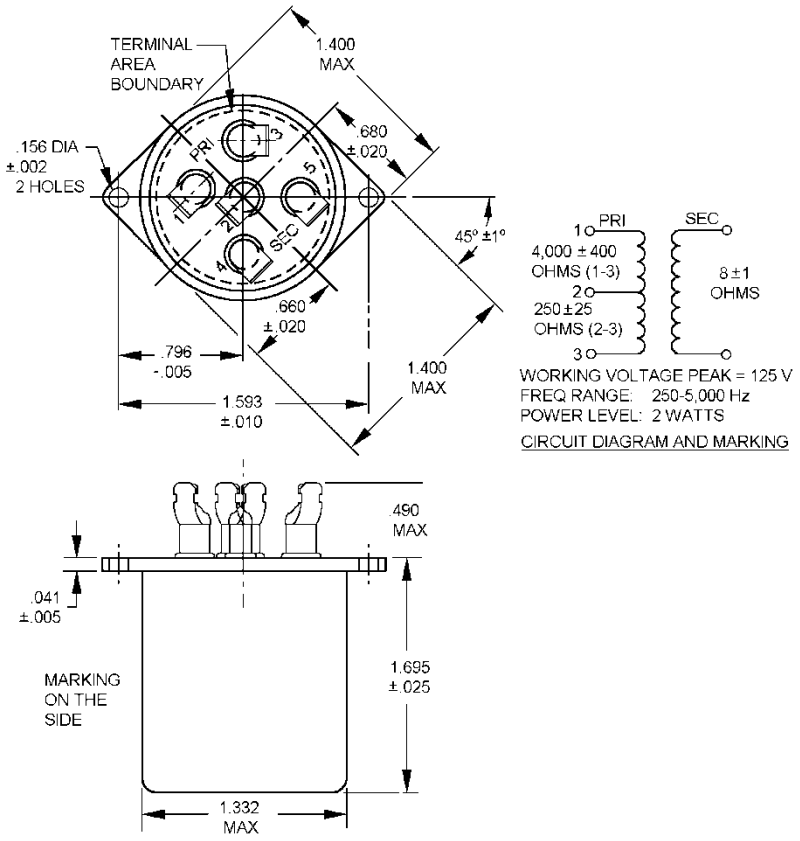
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY

This specification is approved for use by all Departments and agencies of the Department of Defense.

The complete requirements for procuring the product described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

Part number M27/1-01 is inactive for new design after 4 April 1985. For new design use M27/1-02.



Inches	mm
.003	0.08
.005	0.13
.010	0.25
.020	0.51
.025	0.64
.041	1.04
.156	3.98
.490	12.45
.680	17.27
.796	20.22
1.200	30.48
1.332	33.83
1.400	35.56
1.593	40.46
1.695	43.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical rating:

Primary power level: 2 W at 1 kHz.

Working voltage (peak): 125 volts.

Design, configuration, and physical dimensions: See figure 1.

Material: Metal encased, steel.

Weight: 4 ounces, maximum.

Duty cycle: Continuous.

Terminal: Solder lug, number 18 AWG, standoff type.

Temperature range: See table I.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage, 500 V rms.

Electrical characteristics:

Harmonic distortion: Not more than 1 percent at 2 W ±1 dB at 1 kHz.

Primary impedance: 4,000 ±400 ohms (1-3), tapped at 250 ±25 ohms (2-3).

Secondary load impedance: 8 ±1 ohms (4-5).

DC resistance (DCR):

Primary DCR: 157 ohms ±25 percent.

Secondary DCR: 0.56 ohm ±25 percent.

Frequency response:

Range: 250 Hz to 5,000 Hz at rated power, flat within ±2 dB; reference frequency 1 kHz; Test voltage 88.5 V rms (1-3).

Insertion loss: 0.5 dB maximum, with 2 W input at 1 kHz.

Polarity: Additive, with terminals 3 and 4 connected.

Temperature rise: See table I.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B.

Shock (specified pulse): Method 213 of MIL-STD-202. test condition H.

Part or identifying number (PIN): M27/1-(dash number from table I).

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TABLE I. Temperature characteristics.

Dash Number	Temperature range	Maximum operating temperature With rated power at 300 Hz	
		Temperature rise	Ambient temperature
-01	-55°C to +105°C	40°C maximum	65°C
-02	-55°C to +130°C	40°C maximum	90°C

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy – EC
Air Force – 85
DLA – CC

Preparing Activity:
DLA – CC

(Project 5950-2009-092)

Review activities:
Army – AR, CR4
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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